Heat Transfer HW 3

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Submission: 2/24/26

PSET #3

Due: 2/24/26

Worked with
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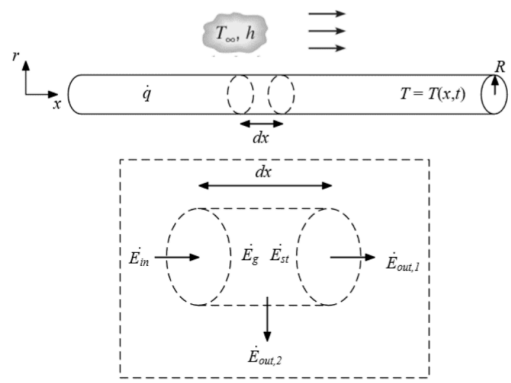
1. **Known:** rad = R, Vol heat gen = $\dot{q}$, T_∞, h, $T = T(x,t)$, k, ρ, c_p, length = L

Unknowns: a. energy terms as function of T, b. governing eqtn, c. boundary condition eqtns

Assumptions: No radiation, radius is small and radial T is uniform, k ρ and c_p constant

Sketch:



Analysis:

a. $\dot{E}_{st} = \dot{E}_{in} + \dot{E}_g - \dot{E}_{out,1} - \dot{E}_{out,2}$

$$\rho c_p V \frac{\partial T}{\partial t} = -kA\frac{\partial T}{\partial x} + V\dot{q} - h(2\pi R)(T - T_\infty)$$

[1] $$\boxed{\rho c_p A\,dx \frac{\partial T}{\partial t} = -kA\frac{\partial T}{\partial x} + A\,dx\,\dot{q} - h(2\pi R\,dx)(T - T_\infty)}$$

b. [1]/dx $$\rho c_p A \frac{\partial T}{\partial t} = -kA\frac{\partial^2 T}{\partial x^2} + A\dot{q} - h(2\pi R)(T - T_\infty)$$

$A = \pi R^2$ /A

$$\rho c_p \frac{\partial T}{\partial t} = -k\frac{\partial^2 T}{\partial x^2} + \dot{q} - \frac{2h}{R}(T - T_\infty)$$

/ρc_p

$$\boxed{\frac{\partial T}{\partial t} = \left(-\frac{k}{\rho c_p}\right)\frac{\partial^2 T}{\partial x^2} + \frac{\dot{q}}{\rho c_p} - \frac{2h}{\rho c_p R}(T - T_\infty)}$$

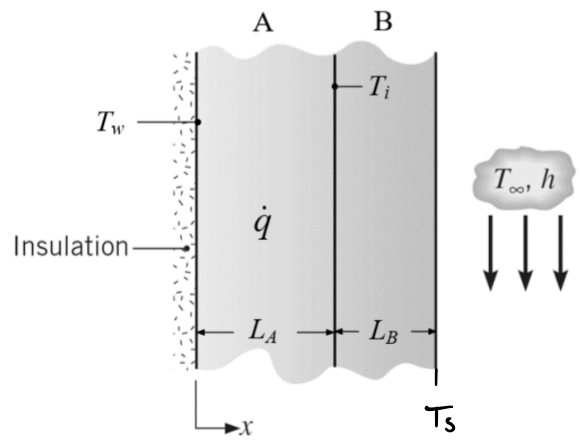
c. $$\boxed{\underset{t>0}{\overset{BC}{T(0,t) = T_1}}, \quad \underset{0 \le x \le L}{\overset{IC}{T(x,0) = T}}, \quad \underset{t>0}{\overset{BC}{-k\frac{\partial T}{\partial x}\Big|_{x=L} = h(T(L,t) - T_\infty)}}}$$

2. Known: $L_A = 0.3m$, $L_B = 0.1m$, $K_B = 30 \frac{W}{mK}$, Vol heat gen$_B = 0$, $T_\infty = 20°C$, $T_w = 115°C$, $T_i = 100°C$, $h = 500\ W/m^2K$

Unknown: a. T_s of wall B, b. $\dot{q}$ steady state, c. K_A, d. temp profile across wall A and B

Assumptions: No heat gen within wall B, A left side isolated, uniform heat flow in x dir

Sketch:



Analysis: a. $q''_{in} = q''_{out}$ (conduction = convection) @ $x = L_A + L_B$ $\rightarrow \frac{K_B(T_i - T_s)}{L_B} = h(T_s - T_\infty) \rightarrow K_B T_i - K_B T_s = hL_B T_s - hL_B T_\infty$

$$\frac{K_B T_i + hL_B T_\infty}{hL_B + K_B} = T_s = \boxed{50°C}$$

b. @ $x = L_A$ $\quad \dot{E}_g = \dot{E}_{out} \rightarrow \dot{q} L_A \cancel{A} = \cancel{A} \frac{K_B(T_i - T_s)}{L_B}$ $\quad \dot{q} = \frac{K_B(T_i - T_s)}{L_A L_B} = \boxed{50kW}$

c. $K_A \frac{d^2T}{dx^2} + \dot{q} = 0 \rightarrow \frac{d^2T}{dx^2} = -\frac{\dot{q}}{K_A} \rightarrow T = -\frac{\dot{q}}{2K_A}x^2 + \cancel{ax}^{0} + \cancel{b}$ @ $x=0$ $T = 115°$ $115°$

$\left.\frac{dT}{dx}\right|_{x=0} = 0 \quad a = 0$ $\quad T = -\frac{\dot{q}}{2K_A}x^2 + 115$

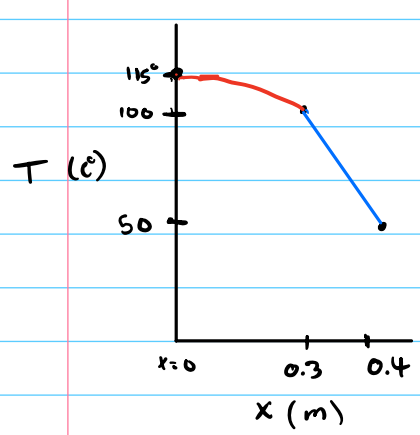
@ $x = L_A$, $T = T_i$

$T_i = -\frac{\dot{q}}{2K_A}L_A^2 + 115$

$\boxed{K_A = 150\ W/mK}$

d. $T = -\frac{\dot{q}}{2K_A}x^2 + 115 \quad 0 < x < L_A$

$L_A < x < L_B$ Linear

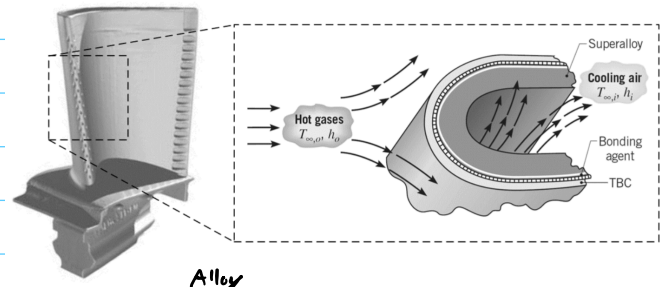


3. **Known:** $k_b = 25\ W/mK$, $k_{TBC} = 1\ W/mK$, $R''_{BA} = 10^{-4}\ m^2K/W$, $T_{\infty,i} = 400K$, $h_i = 500\ W/m^2K$, $T_{\infty,o} = 1616K$, $h_o = 1000\ W/m^2K$, $L_b = 5mm$, $T_{max} = 1200K$

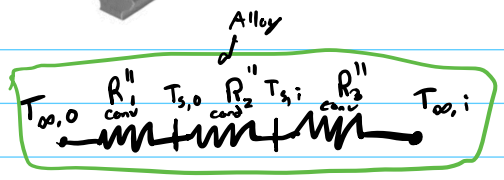
Unknown: a. without TBC can $T_s < T_{max}$, b. with TBC can $T_s < T_{max}$, a. and b. thermal resistance network

Assumptions: steady state conditions, one dim heat transfer through TBC and alloy.

Sketch:



a.



$$R''_{conv} = \frac{1}{h}$$

$$R''_{cond} = \frac{L}{k}$$

$$q'' = \frac{T_{\infty,o} - T_{\infty,i}}{R''_1 + R''_2 + R''_3} = \frac{1616 - 400}{1/1000 + \frac{0.005}{25} + 1/500} = 380{,}000\ W/m^2$$

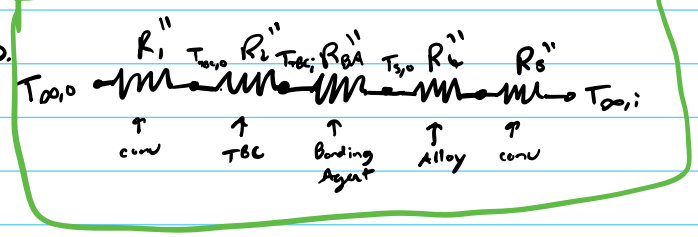
$$q'' = \frac{T_{\infty,o} - T_{s,o}}{R''_1}$$

$$-(3.8\times10^6)(1/1000) + 1616 = T_{s,o}$$

$$\boxed{T_{s,o} = 1236°K}$$

larger than 1200°K so **UNSAFE**

b.



$$q'' = \frac{T_{\infty,o} - T_{\infty,i}}{R''_1 + R''_2 + R''_{BA} + R''_4 + R''_5} = \frac{1616 - 400}{\frac{1}{1000} + \frac{0.0005}{1} + 0.0001 + \frac{0.005}{25} + \frac{1}{500}} = 320{,}000\ W/m^2$$

$= 0.0038$

$$q'' = \frac{T_{\infty,o} - T_{TBC,o}}{R''_1} \rightarrow 1616 - (320{,}000)(0.001) = T_{TBC,o}$$

$1296°K$

$$q'' = \frac{T_{TBC,o} - T_{TBC,i}}{R''_2} \Rightarrow 1296 - (3.2\times10^5)(0.0005) = T_{TBC,i} = 1104°K$$

$$q'' = \frac{T_{TBC,i} - T_{s,o}}{R''_{BA}} \Rightarrow 1104 - (3.2\times10^5)(0.0001) = \boxed{T_{s,o} = 1104°K}$$

less than 1200°K, **SAFE**